



Red Li

About Me

Software & Data Engineer with ~5 years of experience in semiconductor automation and AI. Expertise in Java backend, Python analysis, and ETL optimization. Proven track record in delivering high-efficiency automated solutions.

Contact

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Education

National Chung Hsing University

M.S. in Applied Mathematics

2017-2019

National Chiayi University

B.S. in Applied Mathematics

2013-2017

Skills

- Programming: Java, Python, PL/SQL
- Databases: Oracle DB, MySQL
- Tools/Frameworks: IntelliJ, Weblogic, VS code, Vue.js
- DevOps: Docker, Jenkins, Git, Linux

Certification

Tibame

AI/Big Data Analyst Training Program

2020/04 - 2020/09 (600 hrs)

Experience

Full-Stack Engineer

Macronix International Co., Ltd. **2022/04 - Present**

- Semiconductor Analysis System (Java): Enhancing data processing performance and stability for decision support.
- ETL Automation: Refactored data cleaning workflows (PL/SQL, Java), significantly improving data accuracy and reducing manual effort.
- Frontend Development: Built intuitive interfaces (HTML, Vue.js) for automated material feeding, boosting production line efficiency.

Data Engineer

Winbond Electronics Corp. **2021/04 - 2022/04**

- AI Feature Analysis: Implemented De-noising Auto-Encoders for automated Wafer Map feature extraction and anomaly detection.
- Automated Reporting: Established an automated test data reporting system, replacing manual statistics to improve real-time availability.

Software Engineer

Modovision **2020/10 - 2021/03**

- Computer Vision Pipeline: Built image preprocessing pipelines (OpenCV) and auto-labeling workflows using self-trained models.
- Data Platform: Managed ETL workflows (Airflow, MongoDB to MySQL) and created data visualizations in Tableau.

Project

AI Order Management System (Macronix)

- Award: "Team Excellence Award" Winner.
- Efficiency: Achieved 50% reduction in FT WIP and 70% in AY WIP through precision feeding and automated scheduling.
- Impact: Increased Requested Item Performance (RIP) from 70% to 99%, drastically reducing sales coordination efforts.

AI Production Sales Planning System (Macronix)

- Award: "Team Distinction Award" Winner.
- Responsiveness: Reduced customer inquiry response time to 5 seconds via AI, improving commitment efficiency by 10%.
- Risk Control: Boosted early shipment creation to 96%, enabling balanced stocking and minimizing delivery risks.